

BANCROFT, ON, Canada

WMO: 712940

Lat: 45.0716N

Lon: 77.8799W

Elev: 331

StdP: 97.41

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.7	-25.2	-32.0	0.2	-28.3	-28.5	0.3	-24.5	6.3	-7.8	5.8	-8.6	0.4	0	0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	29.5	20.7	27.9	19.9	26.4	19.0	22.5	26.8	21.5	25.5	20.6	24.3	2.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.2	16.6	24.3	20.3	15.5	23.2	19.3	14.7	22.3	68.0	26.4	64.2	25.7	60.9	24.2	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
5.7	5.1	4.4		-33.8	31.9	2.8	1.6	-35.9	33.1	-37.5	34.0	-39.1	34.8	-41.2	36.0
			DB	-33.6	24.1	3.1	1.2	-35.9	25.0	-37.6	25.7	-39.4	26.3	-41.6	27.2
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.0	-10.3	-8.7	-3.4	4.0	11.3	16.1	18.5	17.4	13.4	6.8	0.1	-6.2		
	DBStd	11.34	7.66	6.62	6.45	4.67	4.50	3.77	3.16	3.20	4.35	4.66	5.32	6.51		
	HDD10.0	2734	629	523	415	188	37	2	0	0	16	122	298	502		
	HDD18.3	4973	888	757	673	431	222	87	38	56	158	357	548	760		
	CDD10.0	908	0	0	1	8	78	186	262	230	117	24	1	0		
	CDD18.3	104	0	0	0	0	4	21	42	28	9	1	0	0		
	CDH23.3	1307	0	0	1	8	104	268	496	313	111	6	0	0		
	CDH26.7	286	0	0	0	1	22	61	117	63	22	0	0	0		
(14) Wind	WSAvg	1.8	1.9	1.9	2.1	2.1	1.9	1.7	1.6	1.4	1.5	1.8	1.9	1.8		
(15) Precipitation	PrecAvg	896	62	55	62	63	79	95	77	82	93	76	76	73		
	PrecMax	1145	115	100	158	121	145	201	136	183	249	139	132	141		
	PrecMin	661	26	19	19	9	29	17	23	38	34	8	32	16		
	PrecStd	127	24	25	36	33	33	50	31	31	51	34	24	34		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.4	8.7	17.7	24.4	29.4	30.4	31.6	31.1	29.4	24.0	15.8	10.8		
		MCWB	7.9	5.0	11.2	14.9	18.8	21.3	22.2	21.5	20.8	17.3	11.0	9.7		
	2%	DB	4.8	5.6	11.8	19.8	26.1	28.5	29.6	28.3	26.4	20.6	12.8	6.9		
		MCWB	4.0	3.5	7.6	11.8	16.8	20.4	21.1	20.2	19.6	15.1	10.8	6.0		
	5%	DB	2.2	2.7	8.6	16.3	23.7	26.6	27.7	26.5	24.1	17.5	10.5	4.2		
		MCWB	1.4	1.3	4.8	9.4	15.1	18.9	19.8	19.3	18.2	14.0	8.4	3.3		
	10%	DB	0.4	1.0	5.7	13.3	21.1	24.4	26.2	24.9	21.7	15.1	8.2	2.0		
		MCWB	-0.3	-0.2	2.8	7.6	14.0	17.7	19.2	18.7	17.3	12.5	6.5	1.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	6.8	13.3	16.5	20.6	23.2	24.7	23.2	22.4	18.9	13.2	10.2		
		MCDB	8.3	7.3	16.8	21.7	26.1	28.0	29.6	27.6	26.4	20.9	14.2	10.7		
	2%	WB	4.1	3.7	8.4	13.5	18.9	21.7	22.7	22.0	20.9	16.5	11.1	6.0		
		MCDB	4.7	5.2	11.1	17.8	24.0	26.4	26.8	25.9	24.6	19.1	12.5	6.7		
	5%	WB	1.6	1.5	5.4	10.5	17.2	20.5	21.5	21.0	19.6	14.6	8.7	3.4		
		MCDB	2.1	2.6	7.9	14.0	21.3	24.4	25.4	24.4	22.6	17.1	10.1	4.1		
	10%	WB	-0.1	0.0	3.1	8.3	15.3	19.1	20.4	20.0	18.2	12.6	6.6	1.3		
		MCDB	0.3	0.9	5.4	12.7	19.2	22.5	24.3	22.9	20.9	14.6	8.0	1.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	12.6	12.3	12.7	14.3	13.0	13.2	13.2	13.5	10.8	9.2	9.4		
		MCDBR	9.8	12.4	15.4	19.4	18.8	15.9	15.3	15.2	15.7	15.2	12.9	9.0		
	5% WB	MCWBR	9.4	10.5	10.5	11.7	9.8	8.3	7.8	8.0	9.1	10.0	10.3	8.3		
		MCWBR	9.6	11.3	13.8	15.5	14.5	13.0	12.5	12.0	13.0	13.4	11.5	8.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.274	0.285	0.305	0.340	0.356	0.396	0.396	0.385	0.358	0.339	0.300	0.270		
		taud	2.375	2.344	2.439	2.431	2.426	2.321	2.331	2.359	2.433	2.483	2.533	2.440		
	Ebn at Noon	Ebn at Noon	846	904	934	922	913	873	867	864	862	832	816	817		
		Edh at Noon	74	92	97	108	112	125	122	115	98	81	65	63		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.35	2.27	3.62	4.49	5.26	5.71	5.79	5.04	3.98	2.34	1.42	1.03		
	RadStd	RadStd	0.16	0.24	0.33	0.43	0.51	0.46	0.45	0.27	0.36	0.22	0.17	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page